



AQUACHECK

Water Potability & Safety Risk Analysis

What actually predicts whether water is safe to drink — and a model built to screen it

3,276

Water samples

9

Physicochemical readings

0.874

Final model test AUC

Can nine water readings tell us if it's safe to drink?

Water utilities, regulators, and field teams need a fast, defensible read on sample safety. This project set out to find what in the data actually predicts potability — and to build a model rigorous enough to trust.



Inputs

9 physicochemical readings: pH, Hardness, Solids, Chloramines, Sulfate, Conductivity, Organic Carbon, Trihalomethanes, Turbidity



Outcome

A binary label — Potable or Not Potable — for each of 3,276 samples



Approach

Test the statistically correct model first; only add complexity if the evidence demands it

A modestly-sized, imbalanced dataset

3,276

total samples

61.0%

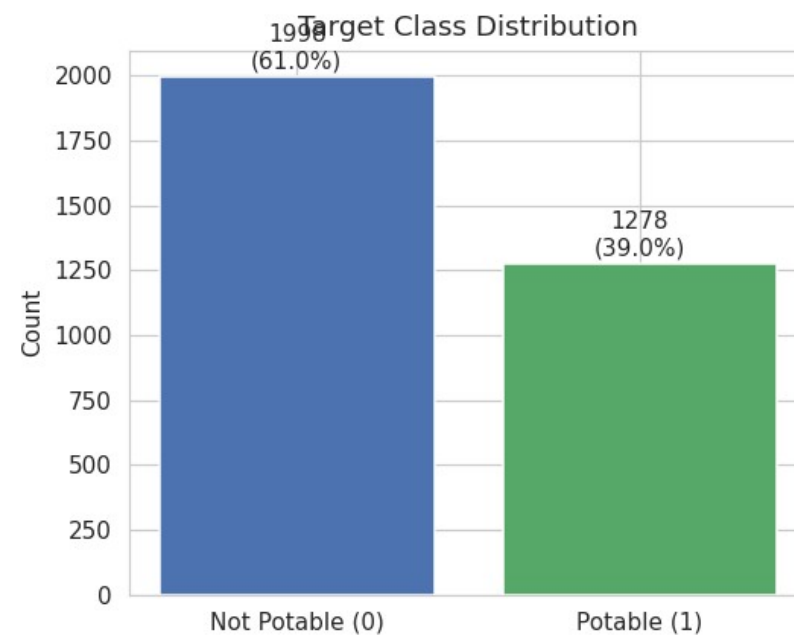
Not potable

39.0%

Potable

A model that always guesses “not potable” is already right 61% of the time — every accuracy number in this project is judged against that baseline.

Target class distribution



Cleaning: filling gaps without hiding the outcome

Three readings had missing values — pH (~15%), Sulfate (~24%), Trihalomethanes (~5%). Before choosing how to fill them, missingness rates were compared between potable and non-potable samples.



Checked missingness pattern

Rates were close between classes on all 3 columns — consistent with missing-at-random, not tied to the outcome.



Imputed within-class median

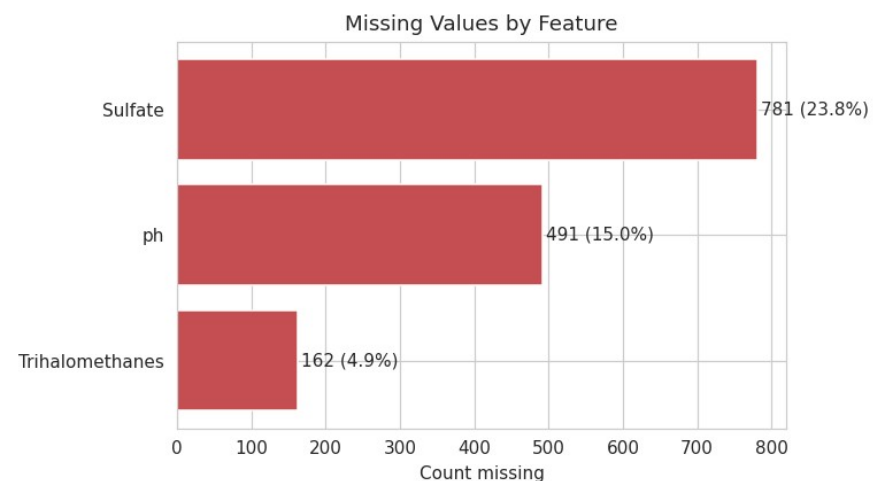
Each missing value filled with its own potability class's median, preserving the class-conditional distribution.



Verified outliers, left them in

IQR scan found under ~3% outliers on every feature — natural sensor spread, not errors. Tree models handle this well.

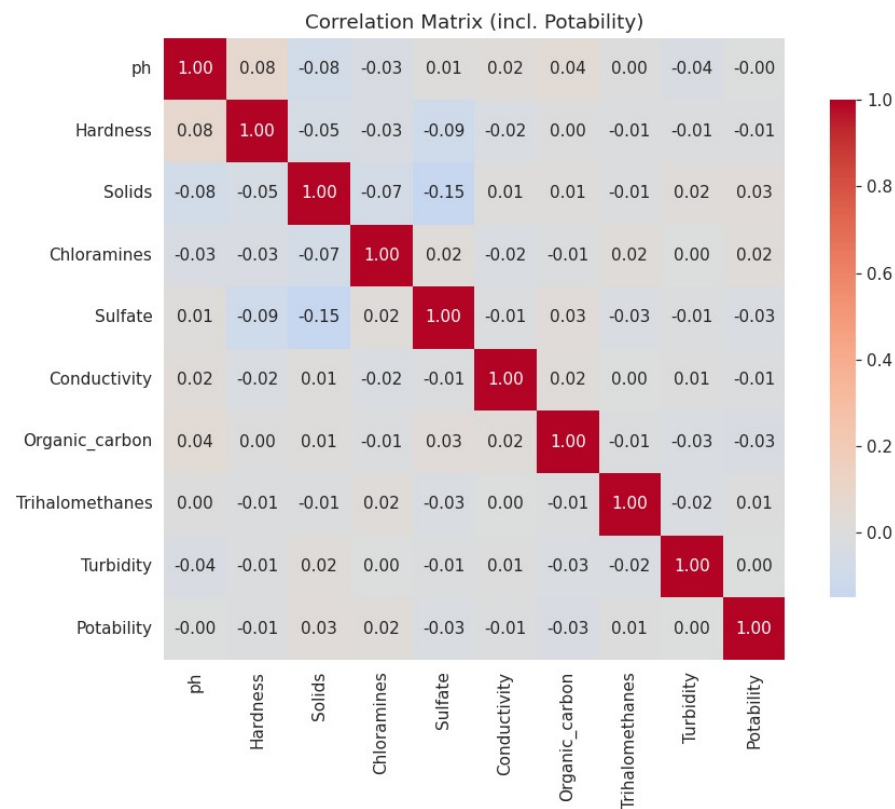
Missing values by feature



0 missing values remain after cleaning.

Every reading overlaps heavily between classes

Correlation matrix



The finding that changes everything

Every single feature's linear correlation with Potability is under 0.04 in magnitude — essentially zero.

That does not mean there's no relationship — it means the relationship, if one exists, is not linear. Section 2 tests that formally; Section 3 shows what a model built to find nonlinear relationships actually finds.

Logistic regression: correct model, no signal to find

The outcome is binary, so OLS is invalid by construction. Logistic regression — the correct linear analogue — was fit and rigorously tested. Four independent diagnostics all point the same direction:

Coefficient significance

Not one of the 9 features significant at $p < 0.05$

Likelihood-ratio test

LR = 4.56, df = 9, $p = 0.87$ — nowhere close to significant

McFadden's pseudo- R^2

0.0014 — over 70× smaller than the weak-fit floor

Hosmer-Lemeshow

$p = 0.035$ — predicted probabilities don't track outcomes



Test AUC: 0.537 — barely above a coin flip (0.500)

This is a valid, important negative result, not a failed analysis: it rules out any single reading — or linear combination of readings — crossing a threshold as the explanation for potability.

If it isn't linear, find a model that isn't either

Tree-based ensembles split the feature space recursively and can represent arbitrary interaction effects — exactly what a near-zero correlation matrix hints is happening here. Three were tuned independently via cross-validated randomized search, then combined into a soft-voting ensemble.



Random Forest

Bagged trees — low variance, easy to reason about



XGBoost

Gradient boosting — typically the strongest single model on tabular data



LightGBM

Gradient boosting — leaf-wise growth, fast and competitive

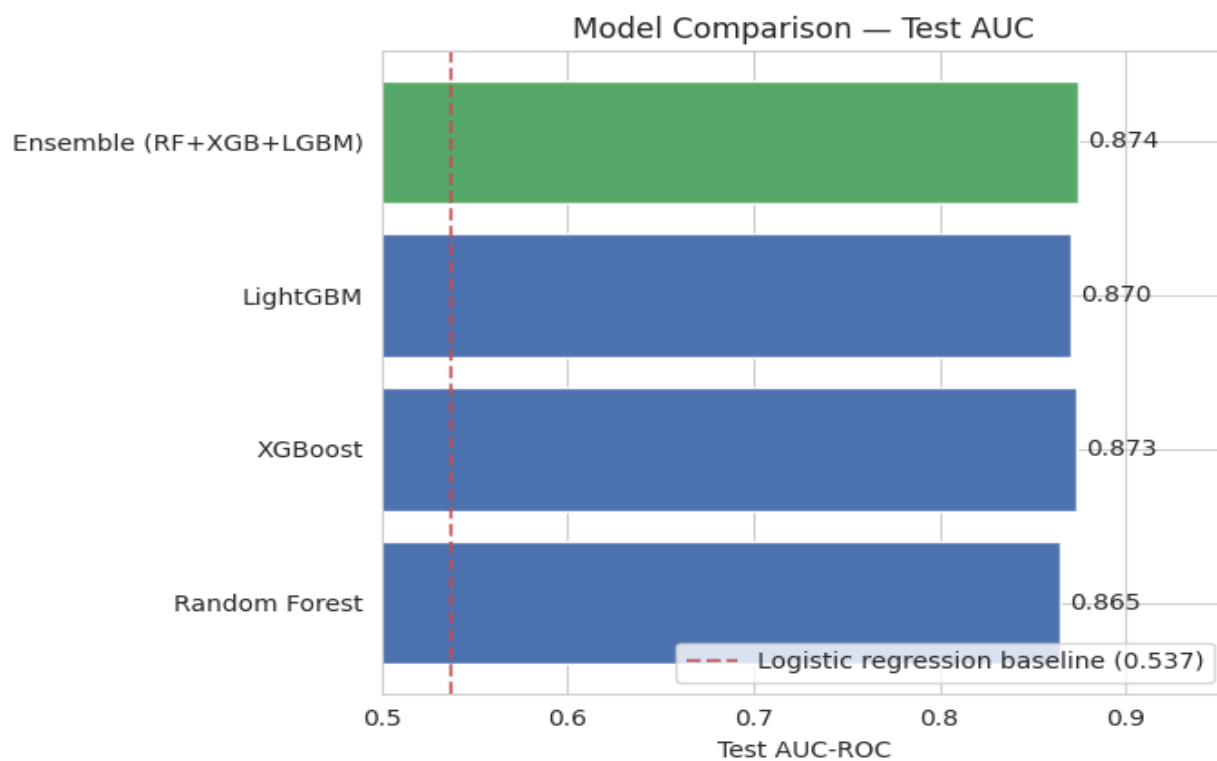


Ensemble

Soft-voting average of all three tuned models

All four candidates comfortably beat the linear baseline

Test AUC by model



FINAL MODEL

Ensemble

RF + XGBoost + LightGBM, soft-voted

0.874

test AUC

OTHER CANDIDATES

Random Forest **0.865**

XGBoost **0.873**

LightGBM **0.870**

Making the model's probabilities trustworthy

Raw ensemble outputs are ranking-optimized, not necessarily well-calibrated — a predicted '70%' doesn't always mean 70% of similar samples really are potable. Isotonic calibration was applied to correct this.

Brier score (raw)

lower is better

0.1387

Brier score (calibrated)

improved

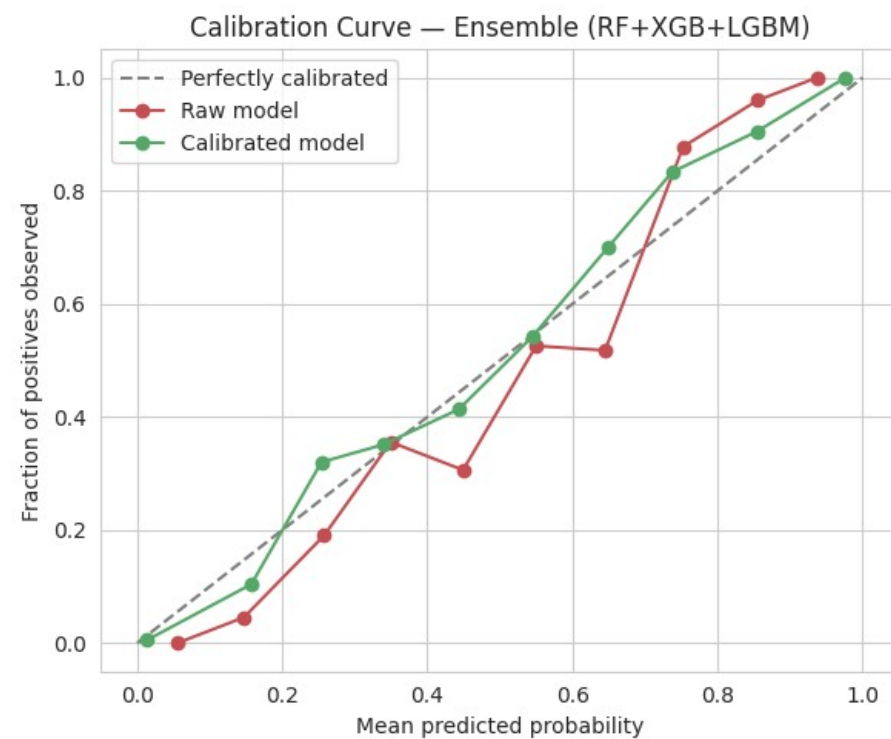
0.1343

AUC change

unaffected — as expected

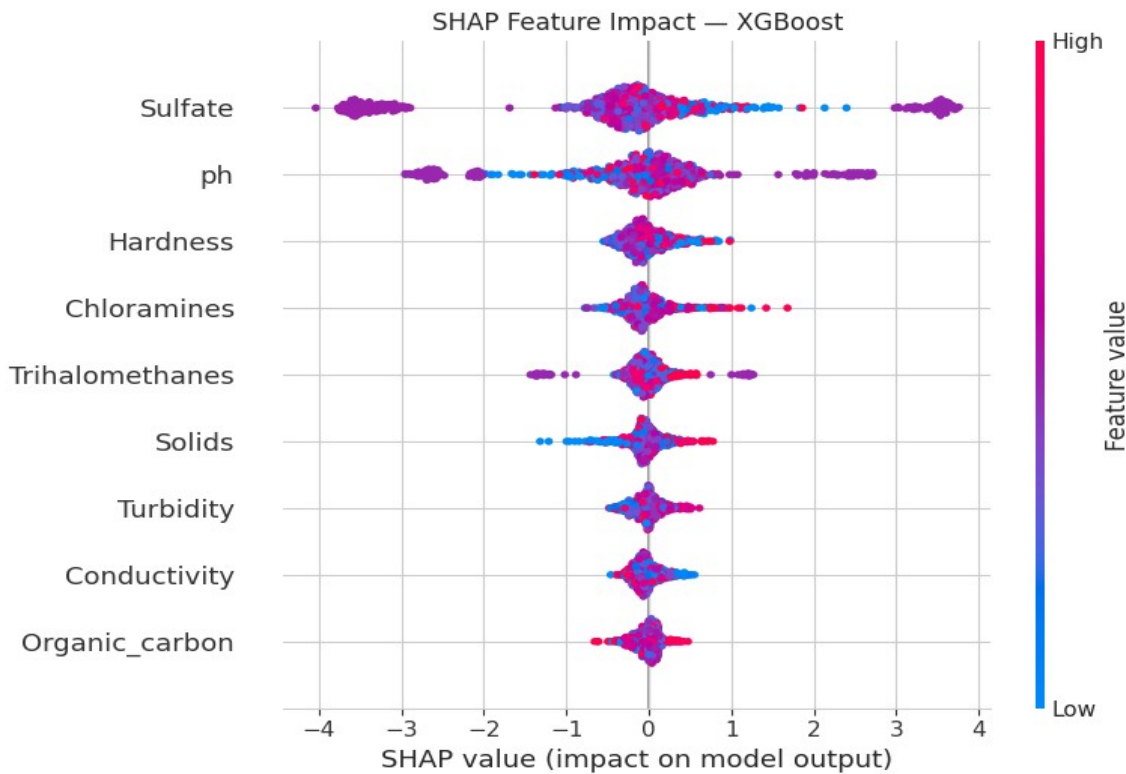
0.874 → 0.874

Calibration curve



What actually drives the prediction

SHAP summary — feature impact & direction



SHAP values quantify how much each reading pushes a prediction up or down, properly accounting for interactions — more rigorous than a tree model’s built-in “importance.”

Sulfate

By far the strongest single driver

pH

Second-strongest, a distant runner-up

Hardness

Third — close to a cluster of secondary features

The model’s decision is genuinely multi-factor — exactly the structure a linear model is mathematically incapable of representing.

What this model can't tell you



No microbiological data

Bacteria, viruses, and parasites are a separate, equally critical safety dimension this model has zero visibility into. A "likely potable" result says nothing about pathogens.



Ceiling on performance

AUC 0.874 is solid, but this dataset is physicochemical-only — there is a real limit on how far any model can go without more signal.



Not locally validated

Trained on this dataset's sample population only. Validate against a specific region's water sources and regulatory standards before operational use.

Key findings and recommendations

KEY FINDINGS



No meaningful linear relationship exists between potability and any of the 9 readings, individually or combined



A tuned tree ensemble captures the true nonlinear relationship: AUC 0.874 vs. 0.537 for the linear baseline



Sulfate and pH are the dominant individual drivers of the model's predictions

RECOMMENDATIONS



Use the tuned ensemble — not logistic regression — for any operational potability screening



Pair any deployment with microbiological testing; this model cannot substitute for it



Validate locally before field deployment — water chemistry varies meaningfully by region